

RaceDay Entity Relationship Diagram

PROG6212 - Programming 2B | PoE Part 1 |

1. Overview

RaceDay is a web-based event management system for South African road running, walking and cycling events. Two roles use the system: Organisers create and manage events, categories and results; Participants browse events, enrol into a category, and view their own results. The data model below uses six entities: **Role**, **User**, **Event**, **Category**, **Enrolment** and **Result**. A single User table stores both Organisers and Participants, distinguished by a RoleID foreign key to a Role lookup table, avoiding duplicated structure between the two account types. Enrolment is an associative entity that resolves the many-to-many relationship between Participants and Events (via Categories).

2. Diagram

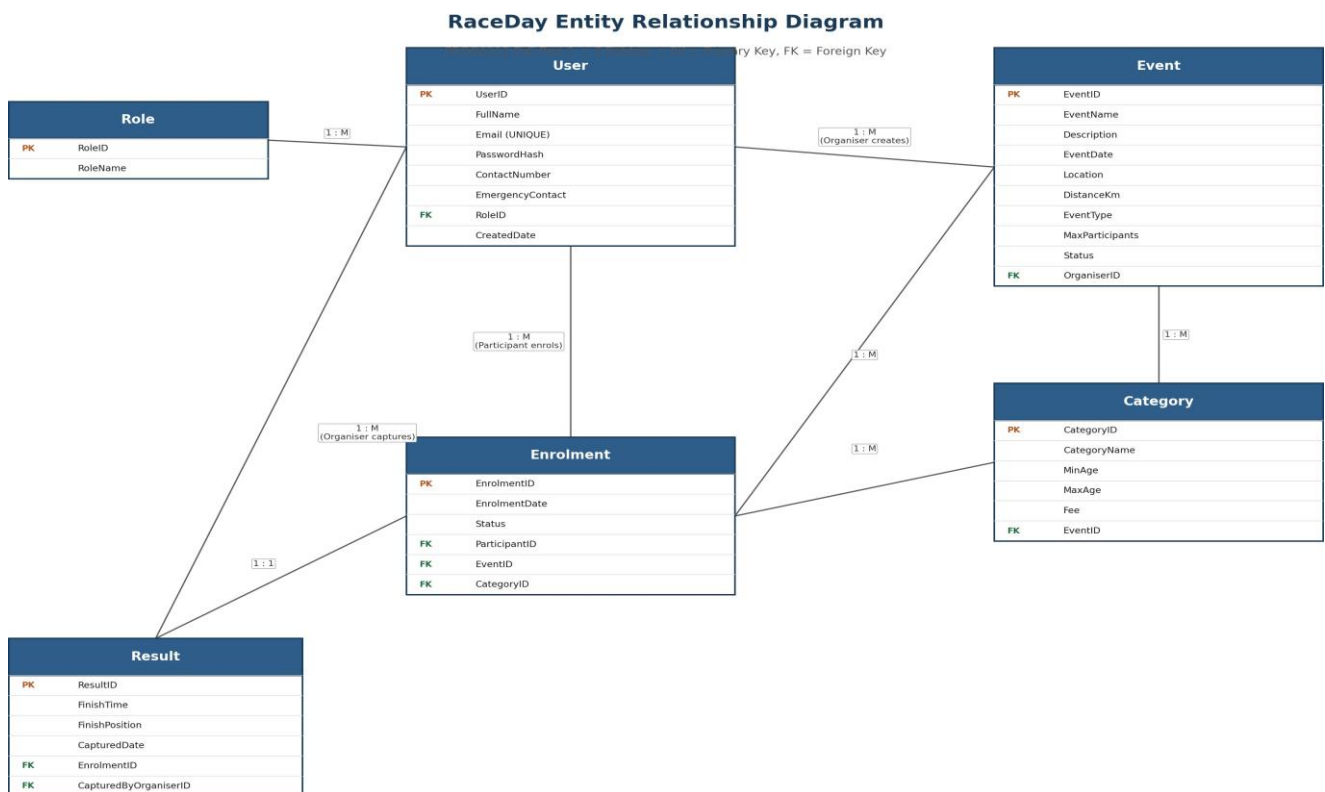


Figure 1: RaceDay ERD showing all entities, attributes, primary keys (PK), foreign keys (FK) and relationship cardinality.

3. Entities and Attributes

3.1 Role

Lookup table distinguishing Organiser accounts from Participant accounts.

Attribute	Key	Data Type / Notes
RoleID	PK	INT, IDENTITY
RoleName		VARCHAR(20), UNIQUE - 'Organiser' or 'Participant'

3.2 User

Stores every RaceDay account. The RoleID foreign key determines whether the account behaves as an Organiser or a Participant.

Attribute	Key	Data Type / Notes
UserID	PK	INT, IDENTITY
FullName		VARCHAR(100), NOT NULL
Email		VARCHAR(150), NOT NULL, UNIQUE
PasswordHash		VARCHAR(255), NOT NULL
ContactNumber		VARCHAR(20)
EmergencyContact		VARCHAR(20) - used mainly for Participants
RoleID	FK	INT, NOT NULL - references Role(RoleID)
CreatedDate		DATETIME, DEFAULT current date/time

3.3 Event

An event created and owned by one Organiser (a User whose RoleID = Organiser).

Attribute	Key	Data Type / Notes
EventID	PK	INT, IDENTITY
EventName		VARCHAR(150), NOT NULL
Description		VARCHAR(500)
EventDate		DATE, NOT NULL
Location		VARCHAR(150), NOT NULL
DistanceKm		DECIMAL(5,2), NOT NULL
EventType		VARCHAR(20) - Running / Walking / Cycling
MaxParticipants		INT, DEFAULT 100
Status		VARCHAR(20), DEFAULT 'Published'
OrganiserID	FK	INT, NOT NULL - references User(UserID)

3.4 Category

A category defined by an Organiser for one specific Event (e.g. age or distance group)

Attribute	Key	Data Type / Notes
CategoryID	PK	INT, IDENTITY
CategoryName		VARCHAR(50), NOT NULL
MinAge		INT, DEFAULT 0
MaxAge		INT, DEFAULT 99
Fee		DECIMAL(8,2), DEFAULT 0
EventID	FK	INT, NOT NULL - references Event(EventID)

3.5 Enrolment

Associative entity resolving the many-to-many relationship between Participants and Events: a Participant enrolls into one Event and selects one Category.

Attribute	Key	Data Type / Notes
EnrolmentID	PK	INT, IDENTITY
EnrolmentDate		DATETIME, DEFAULT current date/time
Status		VARCHAR(20) - Pending / Confirmed / Cancelled
ParticipantID	FK	INT, NOT NULL - references User(UserID)
EventID	FK	INT, NOT NULL - references Event(EventID)
CategoryID	FK	INT, NOT NULL - references Category(CategoryID)

3.6 Result

The finishing outcome for one Enrolment, captured by an Organiser after the event. One-to-one with Enrolment.

Attribute	Key	Data Type / Notes
ResultID	PK	INT, IDENTITY
FinishTime		TIME
FinishPosition		INT
CapturedDate		DATETIME, DEFAULT current date/time
EnrolmentID	FK	INT, NOT NULL, UNIQUE - references Enrolment(EnrolmentID)
CapturedByOrganiserID	FK	INT, NOT NULL - references User(UserID)

4. Relationships and Cardinality

Relationship	Cardinality	Explanation
Role- User	1: Many	One Role (Organiser or Participant) classifies many User accounts
User (Organiser) - Event	1: Many	One Organiser can create many Events; each Event has exactly one owning Other
Event - Category	1: Many	One Event can offer many Categories; each Category belongs to one Event
User (Participant) - Enrolment	1: Many	One Participant can create many Enrolments (enter many events) each enrolment
Event – Enrolment	1: Many	One Event can have many Enrolments; each Enrolment is for one Event
Category - Enrolment	1: Many	One Category can have many Enrolments; each Enrolment selects one category
Enrolment - Result	1: Many	Each Enrolment produces at most one Result once the Organiser captures
User (Organiser) – Result	1: Many	One Organiser can capture many Results; each Result is captured by on Organiser

5. Many-to-Many Resolution

A Participant can enter many Events, and an Event can have many Participants - a natural many-to-many relationship. RaceDay resolves this through the Enrolment associative entity, which sits between User (Participant), Event and Category. This design also naturally supports the requirement that a Participant chooses one Category when enrolling into an Event.

6. Design Notes

- A single User table (rather than separate Organiser/Participant tables) keeps authentication and profile logic simple, while RoleID enforces the distinction required throughout the system.
- Every foreign key traces back to a real functional requirement from the PoE brief (e.g. OrganiserID on Event answers 'how is an Event linked back to its creator?').
- The UNIQUE constraint on Enrolment(ParticipantID, EventID) prevents a Participant from enrolling twice into the same Event.

- This ERD is designed to match the SQL script (RaceDay_Database.sql) and the API Endpoint Plan exactly, table for table and field for field.